Mughesh Kumar N R

Charlottesville, VA | +1-(984)-742-8479 | mugheshkumar9@gmail.com
linkedin.com/in/mughesh | github.com/mughesh03 | mughesh.vercel.app

Education

University of Virginia *Aug 2025 – Dec 2026*
Master of Science in Computer Science Charlottesville, VA
- Advanced Coursework: Computer Networks, Network Engineering, Cyber-Physical Systems, and Distributed Cloud Computing.
- Graduate Teaching Assistant: Mentored and instructed 50+ students in hands-on technical labs, breaking down complex distributed systems, routing protocols, and debugging methodologies into accessible concepts.
- Judged collegiate hackathons and co-led campus engineering ethics initiatives promoting responsible technology adoption.

University of California, Berkeley *Jan 2025 – May 2025*
Startup Semester Program Berkeley, CA
- Completed intensive immersion in Product Management, Technology Go-to-Market Strategy, Customer Discovery, and AI Business Commercialization, formulating solutions for real-world enterprise customer needs.
- Awarded **$10,950** UC Berkeley Chancellor's Scholarship for Startup Semester, selected from 600+ international applicants.

SRM Institute of Science and Technology *Aug 2021 – Dec 2024*
Bachelor of Technology in Computer Science and Engineering (AI & ML) Chennai, India
- Awarded University Gold Medal for Academic Excellence; elected Student Body President representing 20,000+ peers and leading cross-functional campus initiatives alongside university senior leadership.
- Won two international SDG hackathons and **$1,500** in prizes, pitching prototypes to executive industry evaluation juries.
- Published two peer-reviewed IEEE research papers and granted an Indian patent for networked IoT architectures.

Work Experience

University of Virginia Information Technology Services (ITS) *June 2026 – Aug 2026*
Wireless Systems Deployment Summer Intern Charlottesville, VA
- Collaborated with **HPE** network engineers and university infrastructure specialists on a **$6M+ Campus Wireless Refresh** project, modernizing enterprise-grade wireless network connectivity across campus.
- Audited, mapped, and validated deployment configurations, floorplans, and RF coverage metrics for **6,656 Wireless Access Points** utilizing the cloud-native **Juniper Mist AI** network management platform.
- Conducted extensive on-site wireless surveys and physical RF assessments across academic and residential facilities, troubleshooting propagation constraints to deliver seamless end-user service assurance.
- Utilized **ArcGIS** geospatial mapping tools to inventory and audit physical enterprise network hardware, enhancing equipment deployment planning, spatial asset visibility, and cross-team project tracking.

IBM *June 2023 – July 2023*
Artificial Intelligence Intern India
- Engineered automated data analytics and feature-engineering pipelines in **Python (Pandas, NumPy)** on international health datasets to model predictive mental-health Disability-Adjusted Life Years metrics.
- Trained Random Forest predictive models in Scikit-learn, achieving 99.4% R^2 accuracy and outperforming linear baselines.
- Synthesized statistical models and predictive insights into technical documentation, executive presentations, and interactive dashboards, communicating technical trade-offs to cross-functional stakeholders.

Projects

FaceMark AI — Smart Face Recognition Attendance System [Live Demo]
Tech Stack: *JavaScript, WebRTC, Canvas API, Web Speech API, MTCNN, REST APIs, Vercel.*
- Architected a real-time browser-based biometric verification platform utilizing WebRTC video streaming and client-side MTCNN neural networks, extracting 128-D facial feature embeddings for identity matching.
- Implemented interactive 30-sample registration flows, Web Speech voice prompts, 30-minute duplicate cooldown logic, and localStorage persistence with CSV export for real-time auditability and analytics.
- Delivered interactive technical solution demonstrations, validating sub-second inference latency, edge-oriented client compute, and a privacy-first architecture without external server dependencies.

Intelligent Computer Vision System for Pest Detection in Agri Farms
Tech Stack: *Python, TensorFlow, OpenCV, Edge Computing (Jetson Nano), IoT Sensors, Microservices.*
- Designed and deployed an edge-based Computer Vision diagnostic system across regional farms, achieving 96% inference accuracy to automate early pest detection and support sustainable agricultural practices.
- Trained deep convolutional neural networks on 12,000+ field-annotated crop images, classifying 8+ pest species under varying environmental lighting, orientation, and background noise conditions.
- Integrated Python-based microservices with IoT sensor streams and edge hardware (NVIDIA Jetson Nano) for low-latency inference, enabling real-time proof-of-concept telemetry and rapid diagnostic validation.

Technical Skills

Networking & Wireless: HPE Aruba, Juniper Mist AI, Enterprise Wi-Fi, Wireless APs, Site Surveys, TCP/IP, RF Coverage.
Pre-Sales & Solutions: Technical Pre-Sales, Solution Demos, PoC, Requirements Gathering, Client Presentations.
Languages & Tools: Python, C/C++, SQL, JavaScript, HTML/CSS, Bash/Shell, Linux, Git, Docker, Wireshark, REST APIs, Agile, CI/CD.
Cloud, AI & IoT: AWS (VPC, EC2), GCP, Scikit-learn, TensorFlow, OpenCV, Pandas, NumPy, Edge Computing, LoRaWAN.

Publications & Patents

- Energy Efficiency Optimization in Smart Street Lighting Systems via LoRaWAN IoT – IEEE [Publication]
- Handwritten Digit Recognition Comparative Analysis with CNN & LeNet-5 Models – IEEE [Publication]
- IoT-Based Attendance and Monitoring System for Remote Operations Sites – IP India Granted [Patent]